Activity - 1

1. Medical Image Enhancement

A doctor receives an X-ray image in which the bones are visible, but darker tissues have very poor visibility.

i, I would recommend the Power-Law Transformation for medical X-ray image. This transformation is used to adjust brightness and contrast of image by changing its intensity values. Since darker tissues in X-ray image have poor visibility, a gamma value less than 1 can be used to make these dark regions brighter. Therefore, gamma transformation is suitable for improving the visibility & clarity of X-ray image

ii, 1. In the given X-ray image, bones are visible, but darker tissues have poor visibility.

2. Gamma transformation with a gamma value less than 1 increases the brightness of dark pixels

3. It helps reveal hidden details in darker regions

4. It allows doctors to examine the tissues & surrounding areas more clearly.

iii, • The dark regions become brighter
• The visibility of tissues and other details improves.
• The overall image becomes clearer & easier for doctor to analyze.
• The image becomes clearer and more useful for medical examination.

2. Night Time Surveillance

i, I would recommend log transformation to improve visibility of night-time CCTV image. This transformation is useful when most pixel intensity values are concentrated.

ii, • Log transformation expands range of low-intensity pixel values
• It improves visibility of objects in dark areas.
• Negative transformation reverse intensity values but does not specifically enhance the dark regions.

iii, • The dark parking lot becomes brighter
• Car, people, and other objects become more visible
• Details in shadowy regions become clearer

3. 3D-Print Recognition.

i, I would apply the Contrast Stretching Transformation. It improves visibility of fine ridge details and helps fingerprint recognition System analyze image more accurately.

ii, • It increases contrast b/w fingerprint ridges and valleys.
• It makes the ridges & valleys more distinguishable
• This improves fingerprint clarity & helps identify fingerprint patterns.

iii, • Log transformation enhances details in darker regions
• It expands low-intensity values & compresses high-intensity values & compresses high-intensity values
• It may improve some fingerprint details, but it may not provide the same overall contrast improvement as contrast stretching.